The C++ Standard Template Library

Containers hold data, algorithms work on it, iterators connect them. Container choice, operation costs and iterator invalidation are the parts that matter most.

The big picture



Algorithms never touch containers, only iterator ranges.

One sort works on any container that offers the right iterators.

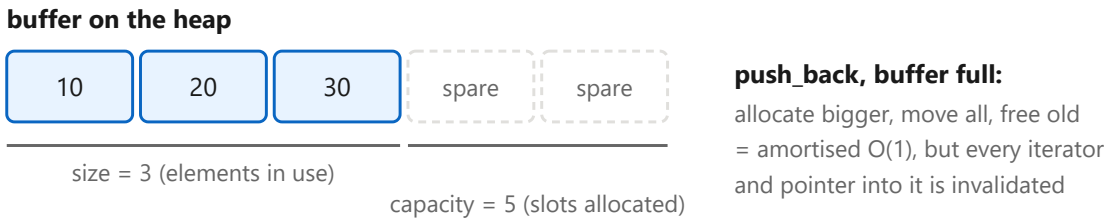
PART	WHAT IT IS	EXAMPLES
Containers	classes that own and organise a collection of elements	`vector`, `map`, `set`, `deque`
Iterators	generalised pointers that walk a container	`v.begin()`, `m.find(k)`
Algorithms	generic functions that operate on iterator *ranges*	`sort`, `find`, `accumulate`

Algorithms see only `[first, last)`, so one `std::sort` serves a vector, array or deque. That separation is the design.

```
#include <vector>       // std::vector
#include <map>          // std::map, std::multimap
#include <unordered_map>// std::unordered_map
#include <set>          // std::set
#include <algorithm>    // std::sort, std::find, std::transform ...
#include <numeric>      // std::accumulate, std::iota
#include <string>       // std::string
```

DEFAULT CHOICE `std::vector`. Contiguous memory is so cache-friendly it often beats "better" containers.

`vector` : the workhorse



reserve() grows capacity only; resize() changes size.

A `vector` is one contiguous heap array with a **size** and a **capacity**. When full, `push_back` allocates a bigger buffer, moves everything and frees the old one: occasional O(n), **amortised O(1)**.

```
std::vector<int> v;
v.reserve(100);          // capacity 100, size still 0: no reallocation for
the next 100 push_backs
v.push_back(1);          // copy/move an existing object in
v.emplace_back(2);       // construct the element in place: no temporary
v.resize(10);            // size becomes 10; new elements are value-
initialised (0)

v[3];                    // no bounds check: out-of-range is undefined
behavior
v.at(3);                 // bounds-checked: throws std::out_of_range
v.front(); v.back(); v.data();   // data() is the raw int* for C APIs
v.size(); v.capacity(); v.empty();
```

RESERVE VS RESIZE `reserve(n)` only allocates; `resize(n)` creates elements. `v[5] = x` after only `reserve(10)` is UB.

BRACES VS PARENTHESES

```
std::vector<int> a(3, 7);   // parentheses: size 3, every element 7  -
> {7, 7, 7}
std::vector<int> b{3, 7};   // braces: the initializer_list constructor
wins  →  {3, 7}
```

With braces, a constructor taking `std::initializer_list` is preferred whenever it can match.

Iterators

An iterator acts like a pointer: `*it`, `++it`, `it == end`. `end()` is **one past the last**; ranges are half-open `[begin, end)`.

```
std::vector<int> v{1, 2, 3};
for (auto it = v.begin(); it ≠ v.end(); ++it) std::cout << *it;   // use ≠
not <, it works for every container
for (int x : v) std::cout << x;           // range-for is exactly this loop
in disguise
for (auto it = v.rbegin(); it ≠ v.rend(); ++it) { }   // reverse iterators
for (auto it = v.cbegin(); it ≠ v.cend(); ++it) { }   // cbegin/cend: read-
only, *it cannot be modified
auto it2 = std::next(v.begin(), 2);       // advance without assuming random
access
```

CATEGORY	CAN DO	CONTAINERS
Forward	`++`, read (multi-pass)	`forward_list`, unordered containers
Bidirectional	also `--`	`list`, `map`, `set`
Random access	also `it + n`, `it[i]`, `it1 - it2`, `<`	`vector`, `deque`, `array`, `string`

So `std::sort` needs random access and fails on `std::list` (use `list.sort()`).

Iterator adaptors: filling a container

An algorithm like `std::copy` **assigns** through its output iterator, so the destination must already be big enough. An **inserter** turns each assignment into an insertion instead:

ADAPTOR	EACH ASSIGNMENT DOES
`std::back_inserter(c)`	`c.push_back(x)`
`std::front_inserter(c)`	`c.push_front(x)` (so the order reverses; needs `deque` / `list`)
`std::inserter(c, pos)`	`c.insert(pos, x)` (works for `set` and `map`)

```
std::vector<int> src{3, 1, 2}, out;
std::copy(src.begin(), src.end(), std::back_inserter(out));      // out is
{3, 1, 2}: it grew
// std::copy(src.begin(), src.end(), out.begin());               // BUG if
out is empty: writes past the end

std::set<int> sorted;
std::copy(src.begin(), src.end(), std::inserter(sorted, sorted.begin()));
```

`std::reverse_iterator` (what `rbegin()` returns) and `std::move_iterator` (moves instead of copying) are the other two adaptors worth knowing.

Iterator invalidation

Changing a container can **invalidate** iterators, pointers and references; using one is UB.

CONTAINER	OPERATION	WHAT IS INVALIDATED
`vector`	`push_back`, `insert`	**all** if it reallocated; otherwise those at or after the insertion point
`vector`	`erase`	the erased element and everything after it
`deque`	insert/erase in the middle	everything; push at either end invalidates iterators but not references
`list`, `map`, `set`	insert	nothing
`list`, `map`, `set`	erase	only the erased element
`unordered_*`	insert that triggers a rehash	all iterators (references stay valid)

▼ SPOT THE BUG **Erasing inside a loop**

```cpp
std::vector<int> v{1, 2, 3, 4};
for (auto it = v.begin(); it ≠ v.end(); ++it)
    if (*it % 2 == 0) v.erase(it);          // BUG
```

`erase` invalidates `it`, and the loop then does `++it` on a dead iterator. The correct idiom uses the iterator `erase` returns:

```cpp
for (auto it = v.begin(); it ≠ v.end(); ) {
    if (*it % 2 == 0) it = v.erase(it);     // erase returns the next
valid iterator
    else ++it;
}
// or, in C++20:
std::erase_if(v, [](int x) { return x % 2 == 0; });
```

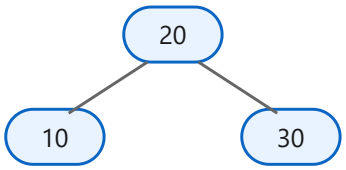
Other sequence containers

CONTAINER	STORAGE	BEST AT	WATCH OUT
`std::array<T,N>`	fixed size, inline (stack)	size known at compile time; zero overhead	cannot grow
`std::vector<T>`	one contiguous heap block	almost everything; fast iteration and indexing	insert/erase in the middle is O(n)
`std::deque<T>`	chunks of arrays	fast push/pop at **both** ends, still indexable	not contiguous
`std::list<T>`	doubly linked nodes	O(1) insert/erase anywhere *if you hold an iterator*; iterators stay valid	no indexing, a heap allocation per element, poor cache behaviour
`std::forward_list<T>`	singly linked nodes	minimal memory list	forward iteration only

OPERATION	VECTOR	DEQUE	LIST
index `[i]`	O(1)	O(1)	not available
`push_back`	amortised O(1)	O(1)	O(1)
`push_front`	O(n)	O(1)	O(1)
insert / erase in the middle	O(n)	O(n)	O(1) at a known position
find a value	O(n)	O(n)	O(n)

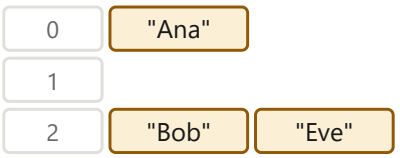
Associative and unordered containers

map / set: balanced tree, sorted, O(log n)



in-order walk = sorted order

unordered_map: hash table, average O(1)



hash(key) picks a bucket; collisions chain (Bob, Eve)

Need order or range queries: map. Plain lookup: unordered_map.

	`MAP` / `SET`	`UNORDERED_MAP` / `UNORDERED_SET`
Implementation	balanced binary tree (usually red-black)	hash table
Order of iteration	**sorted** by key	unspecified
find / insert / erase	O(log n)	average O(1), worst case O(n)
Key requirement	`operator<` (a strict weak ordering)	a hash function and `operator==`
Range queries (`lower_bound`)	yes	no

```cpp
std::map<std::string, int> age;
age["Ana"] = 30;                          // operator[] INSERTS a default (0)
if the key is missing
age.insert({"Bob", 25});                  // does nothing if "Bob" already
exists
age.insert_or_assign("Bob", 26);          // C++17: insert or overwrite

if (auto it = age.find("Ana"); it ≠ age.end())   // find never inserts
    std::cout << it→second;                       // it→first is the key,
it→second the value

for (const auto& [name, years] : age)     // structured bindings; sorted by
name
    std::cout << name << ' ' << years << '\n';

std::unordered_set<int> seen;
if (!seen.insert(5).second) { /* second == false means it was already there
*/ }
```

`try_emplace`, `multimap` and `multiset`

CALL	IF THE KEY ALREADY EXISTS
`m[k] = v`	overwrites (and inserts a default first if missing)
`m.insert({k, v})` / `m.emplace(k, v)`	does nothing, but **may already have moved from or built the value**
`m.try_emplace(k, args ... )` (C++17)	does nothing and **guarantees the arguments are not touched**: compiled, a moved-in 40-character string was intact afterwards
`m.insert_or_assign(k, v)` (C++17)	overwrites

```cpp
std::map<int, std::string> a{{1, "old"}};
std::string payload(40, 'p');
a.try_emplace(1, std::move(payload));     // key 1 exists: nothing happens, payload still holds 40 chars
a.try_emplace(2, std::move(payload));     // new key: the string is moved into the map

std::multimap<std::string, int> mm{{"fruit", 1}, {"fruit", 2}, {"veg", 9}};
// duplicate keys allowed
auto [lo, hi] = mm.equal_range("fruit");   // the range of all entries with that key
mm.count("fruit");                         // 2: there is no operator[] because a key may have many values
```

- `multimap` and `multiset` allow duplicate keys and keep equal keys in insertion order. There is no `operator[]`.
- `try_emplace` is the right call for "insert if absent" when building the value is expensive or the argument is move-only.

Container adaptors

They wrap another container and restrict the interface: `std::stack` (LIFO: `push/pop/top`), `std::queue` (FIFO: `push/pop/front`) and `std::priority_queue` (a heap: `top()` is the **largest** element by default).

```cpp
std::priority_queue<int> maxHeap;                                  // largest first
std::priority_queue<int, std::vector<int>, std::greater<int>> minHeap;   // smallest first
```

Which container should I use?

I NEED TO...	USE	WHY
iterate over items, access by index, mostly append at the end	`std::vector`	Contiguous memory: fastest iteration and indexing, amortised O(1) `push_back`. This is the default choice.
hold a fixed number of items known at compile time	`std::array`	Lives inline (no heap), knows its size, zero overhead over a C array.
push and pop at both ends (a deque of work)	`std::deque`	O(1) push/pop at front and back while still allowing indexing.
insert/erase often in the middle while holding iterators that must stay valid	`std::list`	O(1) splice/insert/erase at a known position and stable iterators. Costs: a heap allocation per node and poor cache locality. Measure before choosing it.
look up values by key; order does not matter	`std::unordered_map`	Hash table: average O(1) find/insert/erase. Needs a hash and `operator==`.
look up by key AND iterate in sorted order, or find the nearest key	`std::map`	Balanced tree: O(log n), sorted iteration, `lower_bound` / `upper_bound` for range queries.
check membership / remove duplicates (no values attached)	`std::unordered_set`	Hash set: average O(1) contains. Use `std::set` if you also need the elements sorted.
count how often each word occurs	`std::unordered_map<std::string, int>`	`counts[word]++` inserts a zero on first sight and increments in one expression.
last-in first-out (undo, DFS, matching brackets)	`std::stack`	Adaptor over deque/vector exposing only push, pop and top.
first-in first-out (task queue, BFS)	`std::queue`	Adaptor over deque exposing push, pop, front and back.
always process the largest (or smallest) item next	`std::priority_queue`	A binary heap: O(log n) push/pop, O(1) top. Largest first by default; pass `std::greater` for smallest first.

KEY POINT "vector vs list?" Answer with *memory*: contiguous means cache-friendly, so vector wins even for middle inserts at modest sizes. List wins for stable iterators and cheap splicing.

Custom keys: comparators and hashes

A `map` / `set` needs an **ordering**; an `unordered_map` / `unordered_set` needs a **hash and equality**. For your own key types, you provide them.

```
struct NoCase {                                          // an ordering that
ignores case
    bool operator()(const std::string& a, const std::string& b) const {
        return std::lexicographical_compare(a.begin(), a.end(), b.begin(),
b.end(),
            [](unsigned char x, unsigned char y) { return std::tolower(x) <
std::tolower(y); });
    }
};
std::map<std::string, int, NoCase> m;                    // the comparator is
part of the TYPE
m["Alpha"] = 1; m["ALPHA"] = 2;                          // the same key:
size is 1 (compiled: 2 keys with "beta")

struct Product { std::string shop; int id; bool operator==(const Product&)
const = default; };
struct ProductHash {
    std::size_t operator()(const Product& p) const {
        std::size_t h = std::hash<std::string>{}(p.shop);
        h ^= std::hash<int>{}(p.id) + 0x9e3779b97f4a7c15ULL + (h << 6) + (h
>> 2);   // combine the member hashes
        return h;
    }
};
std::unordered_map<Product, double, ProductHash> price;
price[{"north", 7}] = 101.5;
```

CONTAINER	KEY MUST PROVIDE	RULE
`map`, `set`	a **strict weak ordering** (`operator<` or a comparator)	irreflexive and transitive; equivalence means "neither is less". `≤` is **not** valid: undefined behaviour
`unordered_map`, `unordered_set`	a hash and `operator==`	equal keys **must** have equal hashes; a poor hash makes every lookup O(n)

- **Transparent comparators:** `std::map<std::string, int, std::less<>>` lets `find` take a `string_view` or `const char*` without building a temporary `std::string` (compiled: it found `"apple"` from a `string_view`). C++20 adds the same for unordered containers.
- **Never change a key** while it is in the container: its position or bucket would be wrong. Keys are `const` for that reason.
- **Hash quality:** combine *all* fields; avoid returning just one field. `std::hash` for `int` is the identity in libstdc++, which is fine.
- A `rehash` (growth beyond `max_load_factor`) invalidates iterators; `reserve(n)` up front avoids it, and avoids the allocations on a hot path.
- Ordered or hashed? Ordered when you need sorted iteration or range queries (`lower_bound`); hashed for plain lookup. For a few dozen keys a sorted `vector` and binary search often beats both.

KEY POINT "How do you use your own type as a key in `unordered_map`?" Provide `operator==` and a hash functor (or specialise `std::hash`), combining all members, and make sure equal objects hash equally.

Algorithms

Over a hundred generic functions in `<algorithm>` / `<numeric>`. Prefer them to loops: named, tested, often faster. Most take a lambda.

```
std::sort(v.begin(), v.end(), [](const P& a, const P& b) { return a.age < b.age; });   // custom order
auto it = std::find_if(v.begin(), v.end(), [](int x) { return x > 6; });
int n  = std::count_if(v.begin(), v.end(), [](int x) { return x % 2 == 0; });
bool any = std::any_of(v.begin(), v.end(), [](int x) { return x < 0; });
int sum  = std::accumulate(v.begin(), v.end(), 0);          // <numeric>
std::transform(v.begin(), v.end(), v.begin(), [](int x) { return x * x; });
auto pos = std::lower_bound(sorted.begin(), sorted.end(), 5);   // binary search: range MUST be sorted
```

STRICT WEAK ORDERING Use `<`, never `≤`: a comparator true for equal elements can crash `sort` or corrupt a `set`. `sort` is not stable; use `stable_sort`.

Ranges and views (C++20)

```
#include <ranges>

std::ranges::sort(v);                                       // pass the container, not begin() and end()
std::ranges::sort(v, std::greater<>{});                     // descending (compiled: first element 6)

auto evenSquares = v | std::views::filter([](int x) { return x % 2 == 0; })
                     | std::views::transform([](int x) { return x * x; });
// a LAZY view: nothing is computed yet
for (int x : evenSquares) use(x);                           // computed on demand, no temporary vector (compiled: sum 56)
```

- A **range** is anything with `begin()` and `end()`. The `std::ranges::` algorithms take the whole range and support **projections**: `std::ranges::sort(people, {}, &Person::age)`.
- A **view** is a lightweight, non-owning, lazy window (`filter`, `transform`, `take`, `drop`, `reverse`, `iota`). Pipe them with `|`.
- A view does not own its data: if the container dies or reallocates, the view dangles, like `string_view`.
- Ranges make pipelines readable; you do not need to know every view.

Common pitfalls

`pair`, `tuple`, `optional` and `variant` are covered in Vocabulary Types.

PITFALL	FIX
Using an iterator/pointer/reference after a `push_back` reallocated the vector	`reserve` up front, or re-fetch the iterator, or index instead
`map[key]` silently inserts	use `find` / `contains` to test; `at()` to read-or-throw
`v.size() - 1` on an empty vector (`size()` is unsigned, so it wraps to a huge number)	check `empty()` first or write `i + 1 < v.size()`
Passing a big container by value	pass `const std::vector<T>&`
`std::vector<bool>` is a packed-bit special case with proxy references	use `std::vector<char>` or `std::bitset`
`auto x = v[0]` copies the element	`const auto& x = v[0]`
`std::find` on a `map` is O(n)	use the member `m.find(key)`, O(log n)

Random numbers: `<random>`

```
std::mt19937 gen(12345);                              // an ENGINE: produces raw
random bits (here seeded for repeatability)
std::uniform_int_distribution<int> dice(1, 6);     // a DISTRIBUTION: shapes
the bits into the numbers you want
int roll = dice(gen);

std::random_device rd;                                // a non-deterministic seed
source (hardware or OS)
std::mt19937 gen2(rd());
std::normal_distribution<double> noise(0.0, 1.0);
```

- **Engine plus distribution**: never use `rand() % n` (poor quality, biased, global state). Same seed gives the **same sequence** (compiled: true), which is what you want in tests and simulations.
- Compiled: 6000 rolls of the dice gave every face between 850 and 1150 times; 10,000 normal samples had a mean near 0.
- Keep one engine per thread; engines are not thread-safe.

Notes: Module 3.

FURTHER READING

C++ Primer (Lippman, Lajoie, Moo)

- 3.3 Library vector type (PDF pages 125 to 134)
- 9.3 Sequential container operations (PDF pages 370 to 383)
- 9.4 How a vector grows (PDF pages 384 to 388)
- 3.4 Introducing iterators (PDF pages 135 to 141)
- 10.4 Revisiting iterators (PDF pages 430 to 438)
- 10.5 Structure of generic algorithms (PDF pages 439 to 443)
- 9.1 to 9.2 Overview and the container library (PDF pages 355 to 369)
- 9.6 Container adaptors (PDF pages 397 to 402)
- 10.6 Container-specific algorithms (PDF pages 444 to 447)
- 11.1 to 11.2 Using an associative container, overview (PDF pages 449 to 456)
- 11.3 Operations on associative containers (PDF pages 457 to 471)

- 11.4 The unordered containers (PDF pages 472 to 477)
- 10.1 to 10.2 Overview and a first look (PDF pages 405 to 413)
- 17.4 Random numbers (PDF pages 774 to 780)

Programming: Principles and Practice (Stroustrup)

- 3.5 to 3.6 Functions; vector introduced (PDF pages 81 to 89)
- 17.6 to 17.9 Other operations, remaining Vector problems, changing size (PDF pages 513 to 523)
- 19.1 to 19.3, 19.6 Containers, sequences and iterators, linked lists, vector/list/string (PDF pages 559 to 572 and 585 to 586)
- 20.1 to 20.6 map, unordered_map, set, container overview (PDF pages 591 to 609)
- 21.1 to 21.2, 21.5 Standard algorithms, function objects, sorting and searching (PDF pages 617 to 626 and 633 to 635)

Effective Modern C++ (Meyers)

- Item 42, Consider emplacement instead of insertion (PDF pages 310 to 320)

C++ Move Semantics (Josuttis)

- Chapter 15 Move semantics in standard library types (PDF pages 229 to 250)
- Chapter 14 Moving algorithms and iterators (PDF pages 219 to 228)